

Acceleration

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

QUESTION		ANSWER	✓
1	State the standard unit of acceleration.	Metre per second squared (m/s^2)	☐
2	State what the gradient of a velocity–time graph represents.	Acceleration	☐
3	State what the area under a velocity–time graph represents.	The distance travelled	☐
4	State the acceleration of an object in free fall near the Earth's surface.	About 9.8 m/s^2	☐
5	Define terminal velocity.	The constant velocity of a falling object when the resistive forces equal its weight.	☐
6	State the equation linking acceleration, change in velocity and time taken.	acceleration = change in velocity $\div$ time taken ($a = \Delta v \div t$)	☐
7	State the equation for uniform acceleration.	$v^2 - u^2 = 2 a s$ (final velocity ² – initial velocity ² = 2 $\times$ acceleration $\times$ distance)	☐
8	State what a negative acceleration means.	The object is slowing down.	☐
9	State what a horizontal line on a velocity–time graph shows.	Constant velocity.	☐
10	State what a straight, sloping line on a velocity–time graph shows.	Constant acceleration.	☐